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and pressure with $V_i$ and $V_f$ as the total volumes of the reactant and product gases respectively, and $n_i$ and $n_f$ as the number of moles of gaseous reactants and products, then,

For reactants (initial state) :

$$PV_i = n_i RT \text{ --------- (7.14)}$$

For products (final state) :

$$PV_f = n_f RT \text{ --------- (7.15)}$$

(7.15) - (7.14)

$$P(V_f - V_i) = (n_f - n_i) RT$$

$$P\Delta V = \Delta n_{(g)} RT \text{ -------(7.16)}$$

Substituting in 7.16 in 7.12

$$\Delta H = \Delta U + \Delta n_{(g)} RT \text{-------(7.17)}$$

### 7.5.2 Enthalpy Changes for Different Types of Reactions and Phase Transitions:

The heat or enthalpy changes accompanying chemical reactions is expressed in different ways depending on the nature of the reaction. These are discussed below.

#### Standard heat of formation

The standard heat of formation of a compound is defined as "the change in enthalpy that takes place when one mole of a compound is formed from its elements, present in their standard states (298 K and 1 bar pressure)". By convention the standard heat of formation of all elements is assigned a value of zero.

$Fe(s) + S(s) \rightarrow FeS(s)$

$$\Delta H_f^0 = -100.42 \text{ kJ mol}^{-1}$$

$2C(s) + H_2(g) \rightarrow C_2H_2(g)$

$$\Delta H_f^0 = +222.33 \text{ kJ mol}^{-1}$$

$$\frac{1}{2} Cl_2(g) + \frac{1}{2} H_2(g) \rightarrow HCl(g)$$

$$\Delta H_f^0 = -92.4 \text{ kJ mol}^{-1}$$

The standard heats of formation of some compounds are given in Table 7.4.

**Table: 7.4 standard heat of formation of some compounds**

| Substance | $\Delta H_f^0$ (kJ mol$^{-1}$) | Substance | $\Delta H_f^0$ (kJ mol$^{-1}$) |
|---|---|---|---|
| $H_2O(l)$ | −242 | $CH_4(g)$ | -74.85 |
| HCl(g) | −92.4 | $C_2H_6(g)$ | −84.6 |
| HBr(g) | −36.4 | $C_6H_6(g)$ | +49.6 |
| NH3(g) | −46.1 | $C_2H_2(g)$ | +222.33 |
| $CO_2(g)$ | -393.5 | $CH_3OH(l)$ | −239.2 |

### 7.6 Thermochemical Equations:

A thermochemical equation is a balanced stoichiometric chemical equation that includes the enthalpy change (ΔH). The following conventions are adopted in thermochemical equations:

(i) The coefficients in a balanced thermochemical equation refer to number of moles of reactants and products involved in the reaction.